Fermi–Amaldi-Based Local Mixing Function

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I. MAIN TEXT

Consider the exact exchange energy density (closed-shell):

$$e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) = -\frac{1}{4}\int \frac{|\gamma(\mathbf{r};\mathbf{r}')|^2}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}', \tag{1}$$

where

$$\gamma(\mathbf{r},\mathbf{r}') = 2\sum_{i=1}^{N/2}\varphi_i^*(\mathbf{r})\varphi_i(\mathbf{r}') \tag{2}$$

is one-electron reduced density matrix [1, 2], and the Fermi–Amaldi [3, 4] energy density expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N,

$$e_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}) = -\frac{1}{2N}\int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}'. \tag{3}$$

For one-orbital systems (one-electron and two-electron singlet), two quantities are equal for all values of $\mathbf{r}$ [5–7]:

$$e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) = e_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}) \quad N \leq 2 \quad \text{for all } \mathbf{r}. \tag{4}$$

For systems with more than one molecular orbital, the spatial behavior of both quantities needs to be studied closer to establish a relation similar to Eq. (4). This is conveniently done by analyzing the corresponding exchange holes.

The exchange hole is defined via

$$e_{\mathrm{x}}(\mathbf{r}) = \frac{1}{2}\int \frac{\rho(\mathbf{r})\rho_{\mathrm{x}}(\mathbf{r},\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}' \tag{5}$$

leading to expressions for the exact and the Fermi–Amaldi exchange holes, respectively:

$$\rho_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r},\mathbf{r}') = -\frac{1}{2}\frac{|\gamma(\mathbf{r},\mathbf{r}')|^2}{\rho(\mathbf{r})}, \tag{6}$$

$$\rho_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}') = -\frac{1}{N}\rho(\mathbf{r}'). \tag{7}$$

When $N > 2$, we cannot establish an unambiguous inequality between these exchnage holes for all $\mathbf{r}$. Although the Cauchy–Schwarz inequality states that

$$\rho(\mathbf{r})\rho(\mathbf{r}') \geq |\gamma(\mathbf{r},\mathbf{r}')|^2 \quad N > 2 \quad \text{for all } \mathbf{r}, \tag{8}$$

the Fermi–Amaldi prefactor, $1/N$, decays faster than the constant prefactor, $1/2$, of the exact exchange hole. Therefore, there are some ranges of $|\mathbf{r}-\mathbf{r}'|$ dominated by one hole and others dominated by another.

As such, in the limit of infinitesimally small interelectronic distances ($r \to r'$) we have:

$$|\gamma(\mathbf{r},\mathbf{r})|^2 = \rho(\mathbf{r})^2, \tag{9}$$

consequently, the depths of the exchange holes are

$$\rho_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r},\mathbf{r}) = -\frac{\rho(\mathbf{r})}{2} \tag{10}$$

and

$$\rho_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{N} \tag{11}$$

thus, for $N > 2$, the well of the exchange hole is dominated by the exact exchange hole.

The long-range behavior of the molecular orbital [8] is given by

$$\phi_i(\mathbf{r}) \sim \exp\left[-(-2\varepsilon_i)^{1/2}r\right], \tag{12}$$

where ε_i is the corresponding eigenvalue. Thus, in the limit of infinitely large interelectromic distances ($|\mathbf{r}-\mathbf{r}'| = \infty$), the electron density is given by:

$$\rho(\mathbf{r}) \sim \exp\left[-2(-2\varepsilon_i)^{1/2}r\right] \tag{13}$$

and the one-electron reduced density matrix is given by:

$$\gamma(\mathbf{r},\mathbf{r}') \sim \exp\left[-2(-2\varepsilon_i)^{1/2}r\right]\exp\left[-2(-2\varepsilon_i)^{1/2}r'\right]. \tag{14}$$

II. SCRAP

A local hybrid exchange functional is given by

$$E_{\mathrm{x}}^{\mathrm{LH}} = \int \left\{g(\mathbf{r})e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) + [1-g(\mathbf{r})]e_{\mathrm{x}}^{\mathrm{SL}}(\mathbf{r})\right\}d\mathbf{r}, \tag{15}$$

where the integrand is a mixture of two exchange density energies weighed by a local mixing function (LMF), $g(\mathbf{r})$, which is position-dependent. The $e_{\mathrm{x}}^{\mathrm{SL}}$ and $e_{\mathrm{x}}^{\mathrm{exact}}$ quantities are a semi-local and the exact exchange energy densities, respectively.

A particular case of a one-electron limit, i.e., the asymptotic limit ($r \to \infty$) where the exact exchange must cancel out an excessive self-repulsion of the Coulomb potential, is also satisfied:

$$\lim_{r\to\infty}\frac{e_{\mathrm{x}}^{\mathrm{FA}}}{e_{\mathrm{x}}^{\mathrm{exact}}} = 1 \tag{16}$$

since both energy densities approach $-1/2r$ as $r \to \infty$.

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